

CS229 Problem Set -3 : Deep learning & Unsupervised Learning

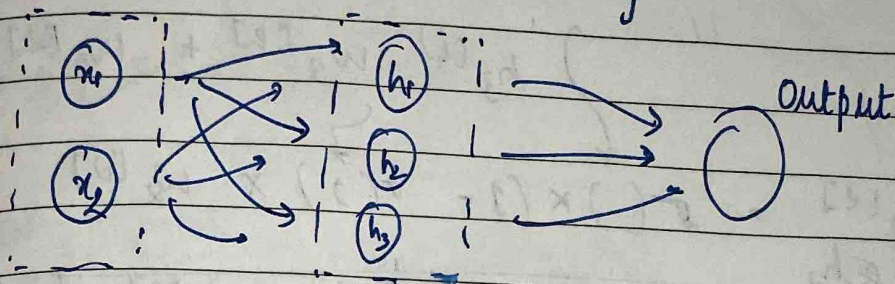
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Q1)

Inputs

Hidden layer



$$x \in \{1, 2\}$$

$$j \in \{1, 2, 3\}$$

Cost / Loss function $\Rightarrow J(\hat{y}, y) = \frac{1}{m} \sum_{i=1}^m L^{(i)}$

$$J(\hat{y}, y) = \frac{1}{m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)})^2$$

where $L^{(i)} = (\hat{y}^{(i)} - y^{(i)})^2$

$$\frac{\partial L}{\partial w} w_{(1,2)}^{[1]} = w_{(1,2)}^{[1]} - \alpha \frac{\partial L}{\partial w_{(1,2)}^{[1]}}$$

$$\frac{\partial L}{\partial w_{(1,2)}^{[1]}} = \frac{\partial L}{\partial a^{[2]}} \cdot \frac{\partial a^{[2]}}{\partial h_2^{[1]}} \cdot \frac{\partial h_2^{[1]}}{\partial w_{(1,2)}^{[1]}}$$

$$L^{(i)} = (\hat{y} - y^{(i)})^2$$

$$L^{(i)} = (\hat{y}^{(i)} - y^{(i)})^2$$

$$L^{(i)} = (a^{[2]} - y^{(i)})^2$$

$$\frac{\partial L}{\partial a^{[2]}} = 2(a^{[2]} - y^{(i)})$$

$\vec{a}$

Also,

$$a^{[2]} = \sigma(h_2^{[1]} \cdot w_{1,2}^{[2]} + h_2^{[1]} \cdot w_{2,2}^{[2]} + h_3^{[1]} \cdot w_{3,2}^{[2]} + w_0^{[2]})$$

$$\frac{\partial a^{[2]}}{\partial h_2} = \sigma(\cdot) \times (1 - \sigma(\cdot)) \times w_{2,2}^{[2]}$$

$$\boxed{\frac{\partial a^{[2]}}{\partial h_2} = a^{[2]} \cdot (1 - a^{[2]}) \times w_{2,2}^{[2]}}$$

Now, we need to calculate $\frac{\partial h_2}{\partial w_{1,2}^{[2]}}$

$$a_2^{[1]} = h_2^{[1]} = \sigma(x_1^{[1]} \times w_{1,2}^{[2]} + x_2^{[1]} \times w_{2,2}^{[2]} + w_{0,2}^{[2]})$$

$$\begin{aligned} \frac{\partial h_2^{[1]}}{\partial w_{1,2}^{[2]}} &= \sigma(\cdot) \times (1 - \sigma(\cdot)) \times x_1^{[1]} \\ &= h_2^{[1]} \cdot (1 - h_2^{[1]}) \times x_1^{[1]} \end{aligned}$$

$$\begin{aligned} \# \frac{\partial L}{\partial w_{1,2}^{[2]}} &= 2(a^{[2]} - y^{[2]}) \times a^{[2]} \times (1 - a^{[2]}) \times \\ &\quad w_{2,2}^{[2]} \times h_2^{[1]} \times (1 - h_2^{[1]}) \times x_1^{[1]} \end{aligned}$$

$$W_{(1,2)}^{[0]} = W_{(1,2)}^{[1]} - \alpha \frac{\partial L}{\partial W_{(1,2)}^{[1]}}$$

~~$$W_{(1,2)}^{[0]} = W_{(1,2)}^{[1]} - \alpha$$~~

~~$$W_{(1,2)}^{[0]} = W_{(1,2)}^{[1]} - \alpha \frac{1}{m} \sum_{i=1}^m \frac{\partial L}{\partial W_{(1,2)}^{[1]}}$$~~

Convert to ~~cost~~ cost function by summing over all m examples.

$$W_{(1,2)}^{[0]} = W_{(1,2)}^{[1]} - \alpha \frac{2W_{(1,2)}^{[1]}}{m} \sum_{i=1}^m (a^{[2]} - y^{[i]}) \times a^{[2]}$$

$$\times (1 - a^{[2]})$$

$$h_2^{[1]} \times (1 - h_2^{[1]}) \times x_1^{[i]}$$

(b) Activation function →

$$f(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$h_1^{[1]} = x_1 w_{1,1} + x_2 w_{2,1} + w_{0,1}^{[1]}$$

$$h_2^{[1]} = x_1 w_{1,2} + x_2 w_{2,2} + w_{0,2}^{[1]}$$

$$h_3^{[1]} = x_1 w_{1,3} + x_2 w_{2,3} + w_{0,3}^{[1]}$$

They form a triangle → If we use step-up activation function, for 3 decision boundaries ~~classified by~~ data will equal to 0 if they are ones inside the triangle

$$x_1 w_{1,1} + x_2 w_{2,1} + w_{0,1} [1] = 0$$

$$x_1 w_{1,2} + x_2 w_{2,2} + w_{0,2} [1] = 0$$

$$x_1 w_{1,3} + x_2 w_{2,3} + w_{0,3} [1] = 0$$

If ~~$x_1 = 0.5$~~ ~~x_2~~
 ~~$w_{1,1} = 0.5$~~ ~~$w_{2,1} = 0$~~ & ~~$w_{0,1} [1] = -1$~~

~~$0.5x_1 + 0 - 1 = 0 \Rightarrow 0.5x_1 = 1$~~
 If $w_{1,1} = 1$, $w_{2,1} = 0$ & $w_{0,1} [1] = -0.5$

~~$x_1 + 0 - 1 = 0$~~

$$x_1 + 0 - 0.5 = 0$$

If $w_{1,2} = 0$, $w_{2,2} = 1$ $w_{0,2} [1] = -0.5$

$$x_2 - 0.5 = 0$$

If $w_{1,3} = 1$ $w_{2,3} = 1$ $w_{0,3} [1] = -4$

$$x_1 + x_2 = 4$$

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Q2) $H(P, Q) = - \sum_{x \in \mathcal{X}} P(x) \log Q(x)$

$$D_{KL}(P, Q) = \sum_{x \in \mathcal{X}} P(x) \log \frac{P(x)}{Q(x)}$$

$$= \sum_{x \in \mathcal{X}} P(x) \log P(x) - \sum_{x \in \mathcal{X}} P(x) \log Q(x)$$

$$= - \sum_{x \in \mathcal{X}} P(x) \log \frac{Q(x)}{P(x)}$$

$$= E \left[- \log \frac{Q(x)}{P(x)} \right]$$

↓ Using Jensen's inequality.

$$\geq - \log E \left[\frac{Q(x)}{P(x)} \right]$$

$$= - \log \left(\sum_{x \in \mathcal{X}} P(x) \frac{Q(x)}{P(x)} \right)$$

$$= - \log \sum_{x \in \mathcal{X}} Q(x)$$

$$= - \log 1 = 0$$

If $P = Q$, then $D_{KL}(P||Q) = \sum_{x \in \mathcal{X}} P(x) \log 1 = 0$

If $D_{KL}(P||Q) = 0$, then $\frac{Q(x)}{P(x)} = E \left[\frac{Q(x)}{P(x)} \right] = 1$

namely $P = Q$

so $D_{KL}(P||Q) = 0$ if $P = Q$.

(b) Chain Rule for KL divergence $\rightarrow$

$$D_{KL}(P(X|Y) || Q(X|Y)) = \sum_y P(y) \left(\sum_x P(x|y) \log \frac{P(x)}{Q(x|y)} \right)$$

To Prove:-

$$D_{KL}(P(X,Y) || Q(X,Y)) = D_{KL}(P(X) || Q(X)) +$$

$$D_{KL}(P(Y|X) || Q(Y|X))$$

$$D_{KL}(P(X,Y) || Q(X,Y)) = \sum_x \sum_y P(x,y) \log \frac{P(x,y)}{Q(x,y)}$$

$$= \sum_x \sum_y P(x) \cdot P(y|x) \log \frac{P(x) P(y|x)}{Q(x) Q(y|x)}$$

$$\sum_x \sum_y P(x) \cdot P(y|x) \left(\log \frac{P(x)}{Q(x)} + \log \frac{P(y|x)}{Q(y|x)} \right)$$

$$= \sum_x \sum_y P(x) \cdot P(y|x) \log \frac{P(x)}{Q(x)} + \sum_x \sum_y P(x) P(y|x) \frac{\log P(y|x)}{\log Q(y|x)}$$

$$= \sum_x P(x) \log \frac{P(x)}{Q(x)} \sum_y P(y|x) + \sum_x P(x) \sum_y P(y|x) \frac{\log P(y|x)}{Q(y|x)}$$

$$= \sum_x P(x) \log \frac{P(x)}{Q(x)} + \sum_x P(x) \left(\sum_y P(y|x) \log \frac{P(y|x)}{Q(y|x)} \right)$$

$$= D_{KL}(P(x) || Q(x)) + D_{KL}(P(y|x) || Q(y|x))$$

$$c) \hat{p}(x) = \frac{1}{m} \sum_{i=1}^m \mathbb{1}\{x^{(i)} = x\}$$

↓
uniform distribution over training set

$$\text{To prove: } \arg \min_{\theta} D_{KL}(\hat{p} \parallel p_{\theta}) = \arg \max_{\theta} \sum_{i=1}^m \log p_{\theta}(x^{(i)})$$

$$D_{KL}(\hat{p} \parallel p_{\theta}) = \sum_{x \in \mathcal{X}} \hat{p}(x) \log \frac{\hat{p}(x)}{p_{\theta}(x)}$$

$$= \sum_{x \in \mathcal{X}} \hat{p}(x) \log \hat{p}(x) - \sum_{x \in \mathcal{X}} \hat{p}(x) \log p_{\theta}(x)$$

$$\arg \min_{\theta} D_{KL}(\hat{p} \parallel p_{\theta}) = \arg \min_{\theta} \sum_{x \in \mathcal{X}} \hat{p}(x) \log \hat{p}(x) -$$

$$\sum_{x \in \mathcal{X}} \hat{p}(x) \log p_{\theta}(x)$$

$$= \arg \max_{\theta} \sum_{x \in \mathcal{X}} \hat{p}(x) \log p_{\theta}(x)$$

$$= \arg \max_{\theta} \sum_{x \in \mathcal{X}} \left(\frac{1}{m} \sum_{i=1}^m \mathbb{1}\{x^{(i)} = x\} \right) \log p_{\theta}(x)$$

$$= \arg \max_{\theta} \sum_{i=1}^m \log p_{\theta}(x^{(i)})$$

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Q3.) KL Divergence, Fisher Information, and the Natural Gradient

$$a) E_{y \sim p(y)} [g(y)] = \int_{-\infty}^{\infty} p(y) g(y) dy$$

To prove:

$$E_{y \sim p(y; \theta)} [\nabla_{\theta'} \log p(y; \theta') |_{\theta' = \theta}] = 0$$

We know:

$$\nabla_{\theta} \log p(y; \theta) = \frac{\nabla_{\theta} p(y; \theta)}{p(y; \theta)}$$

$$E_{y \sim p(y; \theta)} [\nabla_{\theta'} \log p(y; \theta') |_{\theta' = \theta}] = E_{y \sim p(y; \theta)} \left[\frac{\nabla_{\theta} p(y; \theta)}{p(y; \theta)} \right]$$

$$= \int_{-\infty}^{\infty} p(y; \theta) \frac{\nabla_{\theta} p(y; \theta)}{p(y; \theta)} dy$$

$$= \int_{-\infty}^{\infty} \nabla_{\theta} p(y; \theta) dy$$

$$= \nabla_{\theta} \int_{-\infty}^{\infty} p(y; \theta) dy = 0$$

b) $\mathcal{I}(\theta) =$ Fisher information

$$\mathcal{I}'(\theta) = \text{Cov}_{y \sim p(y; \theta)} [\nabla_{\theta'} \log p(y; \theta') |_{\theta' = \theta}]$$

$$\text{Cov}[X] = E[(X - E[X])(X - E[X])^T]$$

$$= E[XX^T] \quad \text{when } E[X] = 0$$

$$\mathcal{I}(\theta) = \text{Cov}_{y \sim p(y; \theta)} [\nabla_{\theta'} \log p(y; \theta') |_{\theta' = \theta}]$$

$$= E_{y \sim p(y; \theta)} [\nabla_{\theta'} \log p(y; \theta') \nabla_{\theta'} \log p(y; \theta')^T |_{\theta' = \theta}]$$

(c) To prove: $E_{y \sim p(y; \theta)} [-\nabla_{\theta'}^2 \log p(y; \theta') |_{\theta' = \theta}] = \mathcal{I}(\theta)$

$$\frac{\partial \log p(y; \theta)}{\partial \theta_i} = \frac{1}{p(y; \theta)} \frac{\partial p(y; \theta)}{\partial \theta_i}$$

$$\mathcal{I}(\theta) = E_{y \sim p(y; \theta)} \left[\nabla_{\theta'} \log p(y; \theta') \nabla_{\theta'} \log p(y; \theta')^T \right]_{\theta' = \theta}$$

$$\mathcal{I}(\theta) = E_{y \sim p(y; \theta)} [\nabla_{\theta'} \log p(y; \theta') \nabla_{\theta'} \log p(y; \theta')^T]_{\theta' = \theta}$$

$$= E_{y \sim p(y; \theta)} \left[\frac{1}{p(y; \theta)^2} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j} \right]$$

$$\frac{\partial^2 \log p(y; \theta)}{\partial \theta_i \partial \theta_j} = -\frac{1}{(p(y; \theta))^2} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j} + \frac{1}{p(y; \theta)} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j}$$

$$E_{y \sim p(y; \theta)} [-\nabla_{\theta}^2 \log p(y; \theta') |_{\theta' = \theta}]_{ij} =$$

$$E_{y \sim p(y; \theta)} \left[\frac{1}{(p(y; \theta))^2} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j} - \frac{1}{p(y; \theta)} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j} \right]$$

$$= E_{y \sim p(y; \theta)} \left[\frac{1}{(p(y; \theta))^2} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j} \right] - E_{y \sim p(y; \theta)} \left[\frac{1}{p(y; \theta)} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j} \right]$$

$$= E_{y \sim p(y; \theta)} \left[\frac{1}{p(y; \theta)^2} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j} \right] - \int_{-\infty}^{\infty} \frac{1}{p(y; \theta)} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j} p(y; \theta) dy$$

$$= E_{y \sim p(y; \theta)} \left[\frac{1}{p(y; \theta)^2} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j} \right] - \frac{\partial^2}{\partial \theta_i \partial \theta_j} \int_{-\infty}^{\infty} p(y; \theta) dy$$

$$= E_{y \sim p(y; \theta)} \left[\frac{1}{p(y; \theta)^2} \frac{\partial^2 p(y; \theta)}{\partial \theta_i \partial \theta_j} \right]$$

$$= \mathcal{I}(\theta)_{ij}$$

a) $\bar{\theta} = \theta + d$

$$\log p(y; \bar{\theta}) \approx \log p(y; \theta) + (\bar{\theta} - \theta)^T \nabla_{\theta} \log p(y; \theta') \Big|_{\theta' = \theta} +$$

$$\frac{1}{2} (\bar{\theta} - \theta)^T (\nabla_{\theta}^2 \log p(y; \theta') \Big|_{\theta' = \theta}) (\bar{\theta} - \theta) =$$

$$\log p(y; \theta) + d^T \nabla_{\theta} \log p(y; \theta') \Big|_{\theta' = \theta} + \frac{1}{2} d^T (\nabla_{\theta}^2 \log p(y; \theta') \Big|_{\theta' = \theta}) d$$

$$\begin{aligned}
 E_{y \sim p(y; \theta)} [\log p(y; \bar{\theta})] &= E_{y \sim p(y; \theta)} [\log p(y; \theta)] + \\
 &\quad \frac{1}{2} d^T E_{y \sim p(y; \theta)} [\nabla_{\theta}^2 \log p(y; \theta')] \Big|_{\theta' = \theta} d \\
 &= E_{y \sim p(y; \theta)} [\log p(y; \theta)] + \frac{1}{2} d^T \mathcal{I}(\theta) d
 \end{aligned}$$

$$\begin{aligned}
 D_{KL}(p_{\theta+d} \| p_{\theta}) &= D_{KL}(p_{\theta} \| p_{\bar{\theta}}) \\
 &= E_{y \sim p(y; \theta)} [\log p(y; \theta)] - E_{y \sim p(y; \theta)} [\log p(y; \bar{\theta})] \\
 &\approx \frac{1}{2} d^T \mathcal{I}(\theta) d
 \end{aligned}$$

b) Newton's method

$$\theta := \theta - H^{-1} \nabla_{\theta} \ell(\theta)$$

$$\mathcal{I}(\theta) = E_{y \sim p(y; \theta)} [-\nabla_{\theta}^2 \log p(y; \theta')] \Big|_{\theta' = \theta}$$

$$= -E_{y \sim p(y; \theta)} [H]$$

$$\theta := \theta + d$$

$$= \theta + \frac{1}{1} \mathcal{I}(\theta)^{-1} \nabla_{\theta} \ell(\theta)$$

$$= \theta - \frac{1}{1} E_{y \sim p(y; \theta)} [H]^{-1} \nabla_{\theta} \ell(\theta)$$

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Q4) Semi-supervised EM.

$$\begin{aligned}
 a) \quad l_{\text{semi-sup}}(\theta^{(t+1)}) &= l_{\text{unsup}}(\theta^{(t+1)}) + \alpha l_{\text{sup}}(\theta^{(t+1)}) \\
 &\geq \sum_{i=1}^m \left(\sum_{z^{(i)}} Q_i^{(t)}(z^{(i)}) \log \frac{p(x^{(i)}, z^{(i)}; \theta^{(t+1)})}{Q_i^{(t)}(z^{(i)})} \right) \\
 &\quad + \alpha \left(\sum_{i=1}^{\bar{m}} \log p(\bar{x}^{(i)}, \bar{z}^{(i)}; \theta^{(t+1)}) \right) \\
 &\geq \sum_{i=1}^m \left(\sum_{z^{(i)}} Q_i^{(t)}(z^{(i)}) \log \frac{p(x^{(i)}, z^{(i)}; \theta^{(t)})}{Q_i^{(t)}(z^{(i)})} \right) \\
 &\quad + \alpha \left(\sum_{i=1}^{\bar{m}} \log p(\bar{x}^{(i)}, \bar{z}^{(i)}; \theta^{(t)}) \right) \\
 &= l_{\text{unsup}}(\theta^{(t)}) + \alpha l_{\text{sup}}(\theta^{(t)})
 \end{aligned}$$

b) Semi-supervised E-step.

Latent variables: $z^{(i)}$

$$w_j^{(i)} = p(z^{(i)} = j | x^{(i)}; \psi, \mu, \Sigma)$$

$$= p(x^{(i)} | z^{(i)} = j; \mu, \Sigma) p(z^{(i)} = j; \phi)$$

$$\sum_{l=1}^K p(x^{(i)} | z^{(i)} = l; \mu, \Sigma) p(z^{(i)} = l; \phi)$$

$$= \frac{1}{(2\pi)^{n/2} |\Sigma_j|^{1/2}} \exp\left(-\frac{1}{2} (x^{(i)} - \mu_j)^T \Sigma_j^{-1} (x^{(i)} - \mu_j)\right)$$

$$\sum_{l=1}^K \frac{1}{(2\pi)^{n/2} |\Sigma_l|^{1/2}} \exp\left(-\frac{1}{2} (x^{(i)} - \mu_l)^T \Sigma_l^{-1} (x^{(i)} - \mu_l)\right)$$